

Paniz Etemad Mazaheri

AUTOMATION & CONTROL SYSTEMS/ ELECTRONICS ENGINEER

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PROFESSIONAL PROFILE

IET Prize-winning Electronic Engineering graduate with two years of hands-on industrial experience as an E, C&I Engineer at Syngenta. Skilled in PLC programming, control system commissioning, and functional safety engineering to IEC 61511. Currently completing an MSc in Robotics and Automation, with applied project work spanning digital twin development, virtual commissioning, and IIoT system integration using Siemens TIA Portal, OPC UA, MQTT, and Node-RED. Competing at national level in WorldSkills UK and the Siemens Xcelator programme. Seeking an engineering role in industrial automation, control systems, or smart manufacturing.

EDUCATION

MSc Robotics and Automation Manchester Metropolitan University	2025 – Sept 2026
BEng (Hons) Electronic Engineering-First Class University of Huddersfield	2021 – 2025
Foundation Year Computing and Engineering University of Huddersfield	2020 – 2021

CORE TECHNICAL SKILLS

- **PLC Programming & Industrial Automation Systems**
TIA Portal, CODESYS, IEC 61131-3 (Ladder Logic, Function Block Diagram, Structured Text), Siemens S7-1200/1500, SCADA systems (WinCC), HMI development
- **Digital Twin Development & Virtual Commissioning**
Siemens NX MCD, Siemens Plant Simulation, CoppeliaSim, model-based systems engineering, virtual commissioning workflows, discrete event simulation
- **Industrial Communication & IIoT Integration**
OPC UA, MQTT, Node-RED, Profibus, Profinet, industrial Ethernet networks, edge computing concepts, Siemens Insights Hub (MindSphere), MES concepts (ISA-95), IEC 61850 (Sampled Values, GOOSE), Hardware-in-the-Loop (HIL) testing, OPAL-RT real-time simulation, IRIG-B time synchronization
- **Functional Safety & Process Safety Engineering**
IEC 61511, IEC 61508, HAZOP, LOPA, SIL assessment using ExSILentia, ATEX compliance, safety lifecycle management, safety instrumented systems (SIS), Lightning as a Source of Ignition
- **Robotics & Control Systems Programming**
ROS (Robot Operating System), MATLAB/Simulink, Python, C++, kinematics & motion control, path planning basics, Siemens SIMATIC Robot Programming Suite, industrial robot integration
- **Electronics & Embedded Hardware Design**
PCB design (schematic & layout), MPLAB X IDE (PIC microcontrollers), CST Studio Suite (EM simulation), embedded system development
- **Engineering Documentation & Systems Tools**
P&ID interpretation, MCC schematics, SRS/FDS documentation, FAT/SAT procedures, Trimble cable calculations

COMPETITIONS & ACHIEVEMENTS

- **WorldSkills UK Industrial Automation:** Finalist in WorldSkills UK Industry 4.0 section.
- **Siemens Xcelerator Programme:** Shortlisted Competitor in Siemens' national engineering competition.

- **IET Prize 2025:** Awarded for Best BEng Electronic Engineering student at the University of Huddersfield.
 - **Best Antenna Design Award:** Designed and validated a VHF Yagi-Uda antenna using CST Studio.
 - **CMI Level 5 Award** in Management and Leadership
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EXPERIENCE

Manchester Metropolitan University

Research Assistant - Smart Grid Lab

2026-Present

- Architected the physical and logical connection between an OPAL-RT simulator and a MiCOM P54 relay for Hardware-in-the-Loop (HIL) testing.
- Configured IEC 61850 Sampled Values (SV) for digitized voltage/current data transfer and GOOSE messaging for bi-directional trip signals.
- Established IRIG-B time synchronization for microsecond-level time coherence.
- Built and ran real-time power system models using MATLAB/Simulink and RT-LAB; configured the relay using MiCOM S1 Agile.

Syngenta

Part-Time Functional Safety Engineer

2024-2025

- Designed an IEC 61511-compliant overfill protection system for a COMAH-regulated solvent storage facility, covering the full safety lifecycle from HAZOP to SIL verification.
- Performed SIL 1 vs SIL 2 comparative analysis using PFDavg calculations and Exida ExSILentia.
- Conducted cost-benefit analysis, demonstrating SIL 1 met all safety targets while avoiding unnecessary system complexity and cost.
- Selected instrumentation based on functional safety performance metrics

Syngenta

E,C&I Engineering Intern

2023-2025

- Supported software commissioning of PCS and PLC systems on active production lines, including logic validation, fault troubleshooting, and functional acceptance testing.
- Programmed and simulated PLC logic in Siemens TIA Portal, contributing to the commissioning of a new production unit.
- Interpreted and updated P&IDs, MCC and engineering documentation during live commissioning.
- Performed on-site electrical measurements and motor frequency analysis using diagnostic tools.
- Gained hands-on exposure to Profibus and Fieldbus communication networks in a COMAH-regulated environment.

University of Huddersfield

Student Ambassador and Student Representative

2022-2025

- Communicated clearly and professionally with students and families in a friendly and respectful manner.
 - Delivered campus tours to prospective students and parents, engaging with diverse age groups.
 - Worked effectively within a large student team to identify and resolve issues quickly and collaboratively.
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KEY ENGINEERING PROJECTS

IoT Virtual Commissioning System

Siemens NX MCD · TIA Portal · OPC UA · MQTT/TLS · Node-RED · PLCSIM Advanced

MSc Project-2025

- Built a virtual commissioning system integrating a Festo CP Lab assembly line with Siemens NX Mechatronics Concept Designer, TIA Portal, and PLCSIM Advanced via OPC UA.
- Implemented a secure IIoT pipeline using MQTT over TLS and Node-RED for real-time monitoring and control.

UR7e Robot Arm - Kinematic Analysis

MATLAB · CoppeliaSim · Peter Corke Robotics Toolbox

MSc Project-2025

- Derived DH parameters from real robot geometry and implemented forward and inverse kinematics in MATLAB using a damped least-squares Jacobian solver.
- Simulated workspace analysis in CoppeliaSim.

IEC 61511 Overfill Protection System

ExSILentia · CAD · IEC 61511 · COMAH

BEng Dissertation-2025

- Developed a standardised SIL design procedure for a solvent storage vessel, from HAZOP and LOPA through to PFDavg calculation and SRS documentation.
- Received First Class honours; project delivered in collaboration with Syngenta.